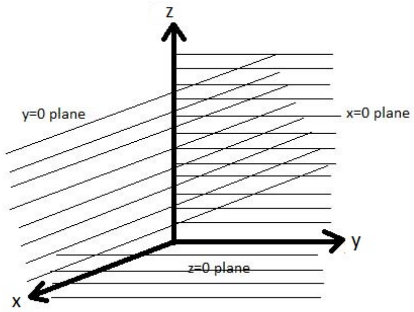
Module 5: MULTIPLE INTEGRATION - II

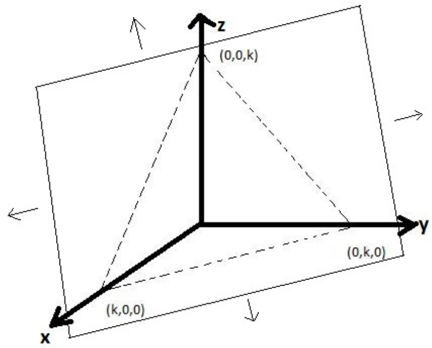
Extension from 2D to 3D

Let us understand how to integrate in a 3-dimensional space by strengthening our basics about geometry in 3D

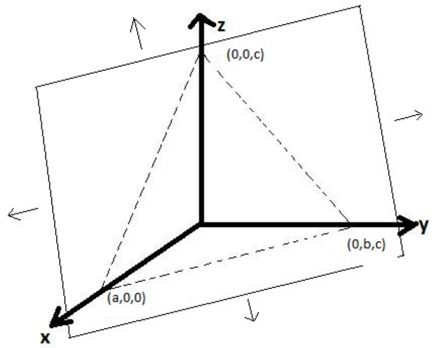
1. Equation $x = 0$ represents the $yz-$ plane in 3D
   Similarly $y = 0$ represents the $xz-$ plane and
   $z = 0$ represents the $xy-$ plane



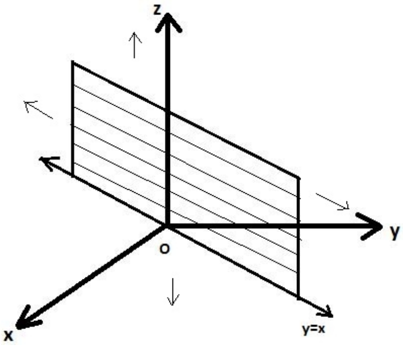
2. $x+y+z = k$ represents a plane passing through the points $(k, 0, 0), (0, k, 0)$ *and* $(0, 0, k)$



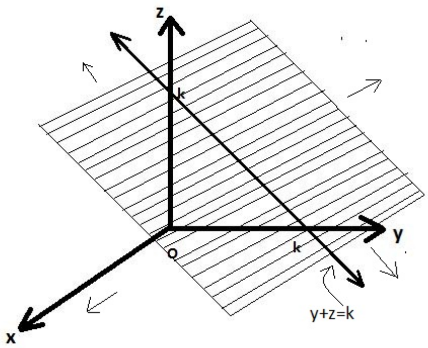
   Similarly $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$ represents a plane passing through the points (a,0,0) , (0,b,0) and (0,0,c)



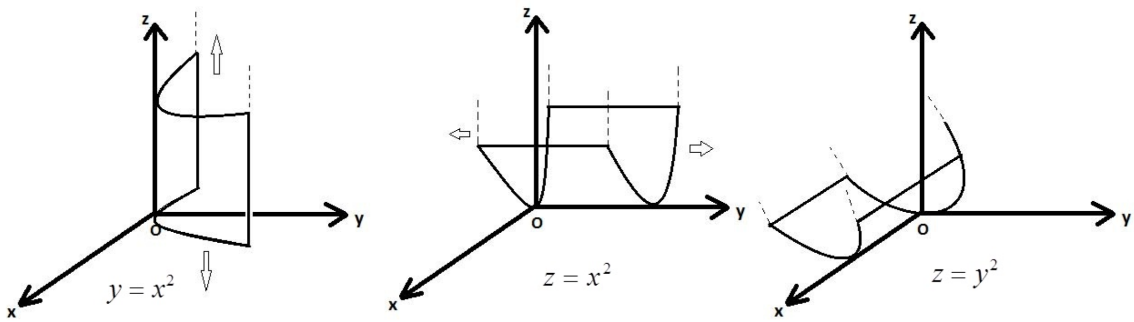
The equation $y = x$ represents a plane passing through origin along $z-$ axis.



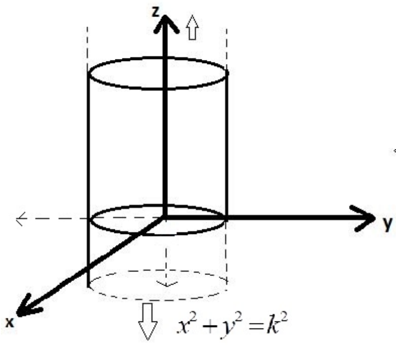
Similarly we have $y + z = k$, etc.

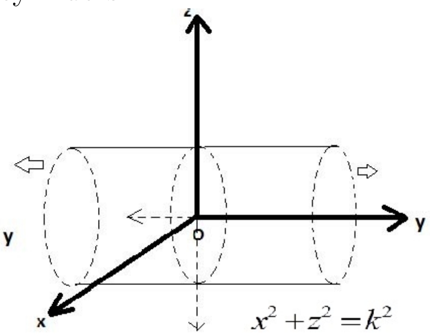


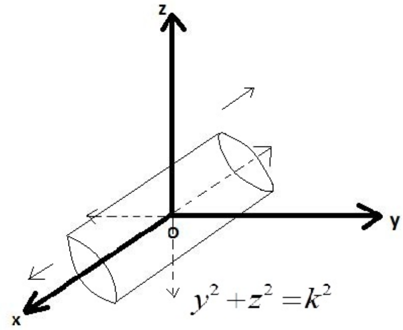
3. $y = x^2$ represents a parabolic cylinder in 3D etc.



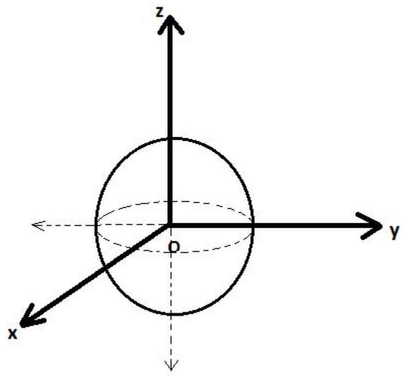
4. $x^2 + y^2 = k^2$ etc represents cylinders.





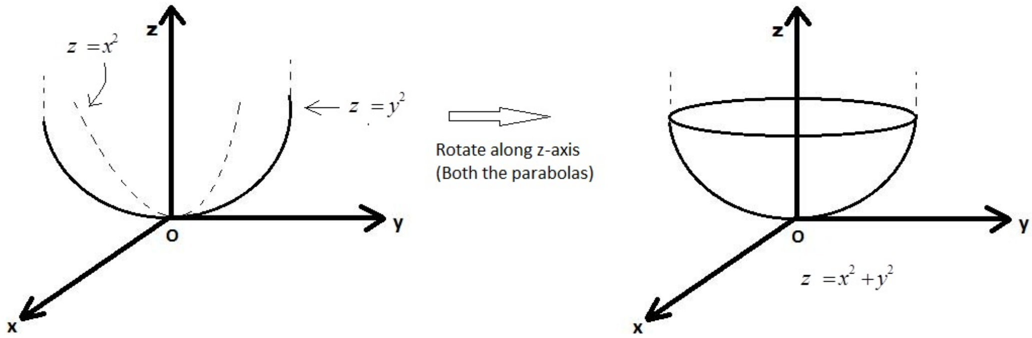


5. Equation $x^2 + y^2 + z^2 = k^2$ represents a sphere centre at origin with radius 'k'.

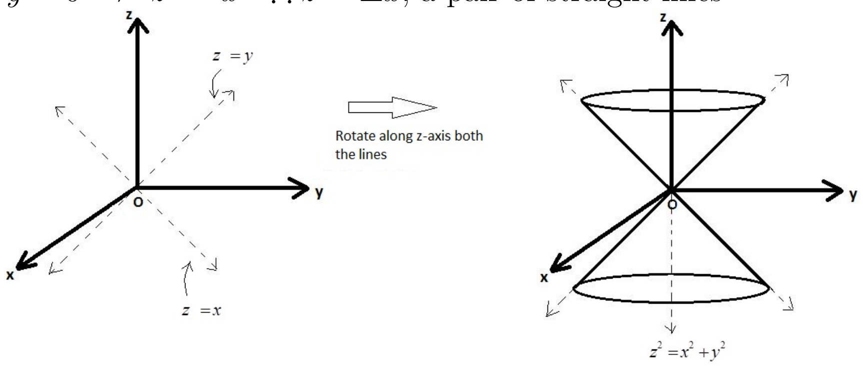


In general, $(x-a)^2+(y-b)^2+(z-c)^2 = r^2$ represents a sphere centre at (a,b,c) with radius 'r'.

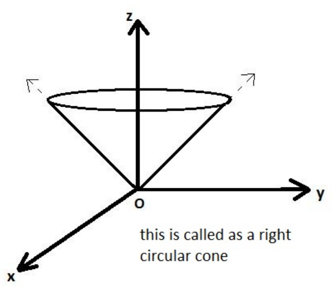
6. $z = x^2 + y^2$ represents an object which is called as a **Paraboloid.**
   Note that $z \geq 0$ (since, $z = x^2 + y^2$ and x and y are real numbers)
   Put y=0, we get $z = x^2$ which is a parabola symmetric about positive z-axis in xz-plane.
   now put x=0, we get $z = y^2$ which is a parabola symmetric about positive z-axis in yz-plane.



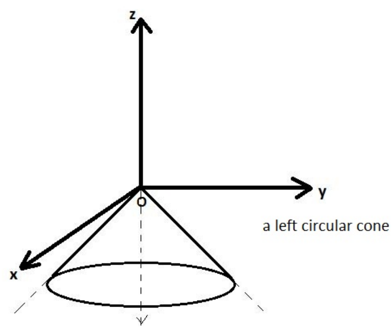
7. $z^2 = x^2 + y^2$ represents a **cone**
   $x = 0 \Rightarrow \ z^2 = y^2 \ \therefore z = \pm y$, a pair of straight lines
   $y = 0 \ \Rightarrow \ z^2 = x^2 \ \therefore z = \pm x$, a pair of straight lines



If $z \geq 0$ then we have



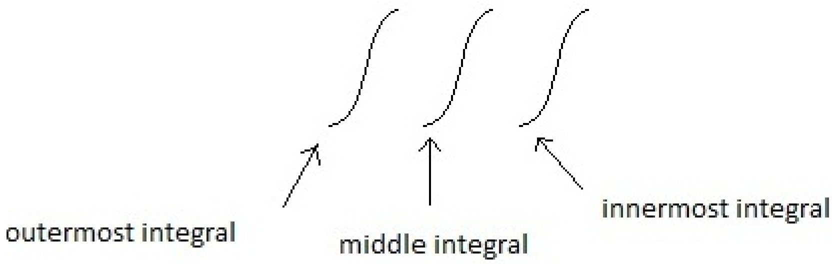
similarly we have if $z \leq 0$ then



1. Evaluate

$$\int_0^2 \int_0^y \int_{x-y}^{x+y} (x+y+z)\, dz\, dx\, dy$$

<u>solution</u>:- In Triple Integration, in this case since inner most integral limits



are in terms of x and y so they have to be z limits. so we integrate w.r.t dz first (treating x and y constant) and middle integral limits are in terms of 'y' so they have to be 'x' limits . so we integrate w.r.t dx(treating 'y' constant) and finally we will integrate w.r.t dy.

Let $I = \int_{y=0}^{y=2} \int_{x=0}^{x=y} \int_{z=x-y}^{z=x+y} (x+y+z)\ dz\ dx\ dy$

By using the formula: $\int (at+b)^n\ dt = \dfrac{(at+b)^{n+1}}{(n+1)a}$

$$\begin{aligned}
I &= \int_0^2 \int_0^y \frac{(x+y+z)^2}{2}\Big|_{z=x-y}^{z=x+y} dxdy \\
I &= \frac{1}{2}\int_0^2 \int_0^y \left\{[(x+y)+(x+y)]^2 - [(x+y)+(x-y)]^2\right\} dxdy \\
I &= \frac{1}{2}\int_0^2 \int_0^y [(2x+2y)^2 - (2x)^2] dxdy = \frac{4}{2}\int_0^2 \int_0^y [(x+y)^2 - (x)^2] dxdy \\
I &= 2\int_0^2 [\frac{(x+y)^3}{3} - \frac{x^3}{3}]\Big|_0^y dy = \frac{2}{3}\int_0^2 \left\{[(y+y)^3 - y^3] - y^3\right\} dy \\
I &= \frac{2}{3}\int_0^2 [8y^3 - 2y^3] dy = \frac{2}{3}\int_0^2 [6y^3] dy \\
I &= \frac{2}{3} \times 6[\frac{y^4}{4}]\Big|_0^2 \\
I &= 4 \times \frac{2^4}{4} = 16
\end{aligned}$$

2. Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{x+y+z} dx\ dy\ dz.$

solution:- Let $I = \int_0^{\log 2}\int_0^{x}\int_0^{x+y} e^{x+y+z}dx\ dy\ dz$ which can be written as,

$$\begin{aligned}
I &= \int_{x=0}^{x=\log 2}\int_{y=0}^{y=x}\int_{z=0}^{z=x+y} e^{x+y}\cdot e^{z} dzdydx \\
&= \int_0^{\log 2}\int_0^{x} e^{x+y}[e^{z}]\Big|_0^{x+y} dydx \\
&= \int_0^{\log 2}\int_0^{x} e^{x+y}[e^{x+y}-e^{0}]dydx \\
&= \int_0^{\log 2}\int_0^{x}[e^{2x+2y}-e^{x+y}]dydx \\
&= \int_0^{\log 2}[e^{2x}\cdot\frac{e^{2y}}{2}-e^{x}\cdot e^{y}]\Big|_0^{x} dx \\
&= \int_0^{\log 2}[e^{2x}\cdot\frac{e^{2x}}{2}-e^{x}\cdot e^{x}-\left(e^{2x}\cdot\frac{1}{2}-e^{x}\right)]dx \\
&= \int_0^{\log 2}[\frac{e^{4x}}{2}-\frac{3}{2}e^{2x}+e^{x}]dx \\
&= \frac{1}{8}\left[e^{4x}\right]\Big|_0^{\log 2}-\frac{3}{4}\left[e^{2x}\right]\Big|_0^{\log 2}+\left[e^{x}\right]\Big|_0^{\log 2} \\
&= \frac{1}{8}\left[e^{4\log 2}-e^{0}\right]-\frac{3}{4}\left[e^{2\log 2}-e^{0}\right]+\left[e^{\log 2}-e^{0}\right] \\
&= \frac{1}{8}\left[e^{\log 2^4}-1\right]-\frac{3}{4}\left[e^{\log 2^2}-1\right]+[2-1] \\
&= \frac{1}{8}[15]-\frac{3}{4}[4-1]+[1] \\
\therefore I &= \frac{5}{8}
\end{aligned}$$

To Evaluate Triple Integration over a given volume

consider a triple integration, $\iiint\limits_V f(x, y, z)\, dz\, dy\, dx$ over a volume 'v'.

step 1: Sketch the volume 'V' along with its "shadow" R (vertical Projection) in the xy-plane.Label the Upper and Lower surfaces of 'V' the bounding curves of R.

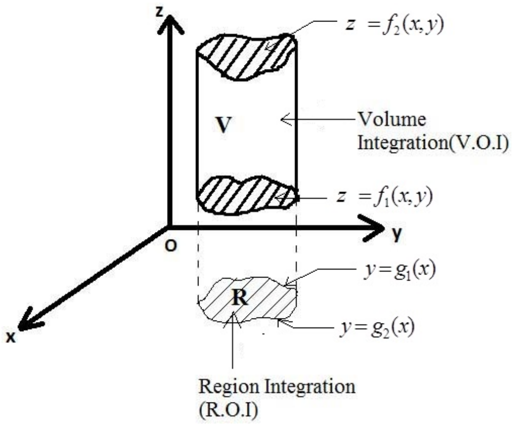


Figure 1:

step 2: To find z-limits of integration draw a line 'l' passing through a volume 'V' parellel to z-axis as z-increases , 'l' enters 'V' at z=$f_1(x, y)$ leaves at z=$f_2(x, y)$. These are the z-limits of integration. step 3: to find 'x' and 'y' limits:-

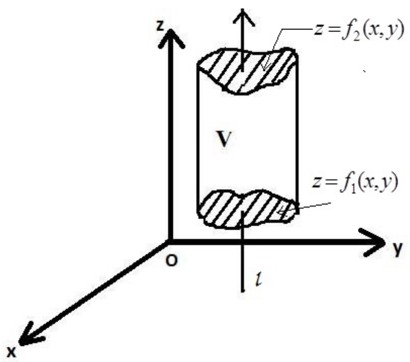


Figure 2:

Consider Region of Integration (R.O.I) in xy-plane use the double integration method to find x and y limits.

<u>Note</u>:- follow similar procedure if you change the order of integration.The shadow of volume of integration. The shadow of volume 'V' lies in the plane of the last two variables with respect to which the integration takes place.

1. Evaluate $\iiint\limits_V \frac{dx\,dy\,dz}{(1+x+y+z)^2}$ over the volume 'V' of the tetrahedron bounded by x=0, y=0, z=0, x+y+z=1 .

   <u>solution</u>:-

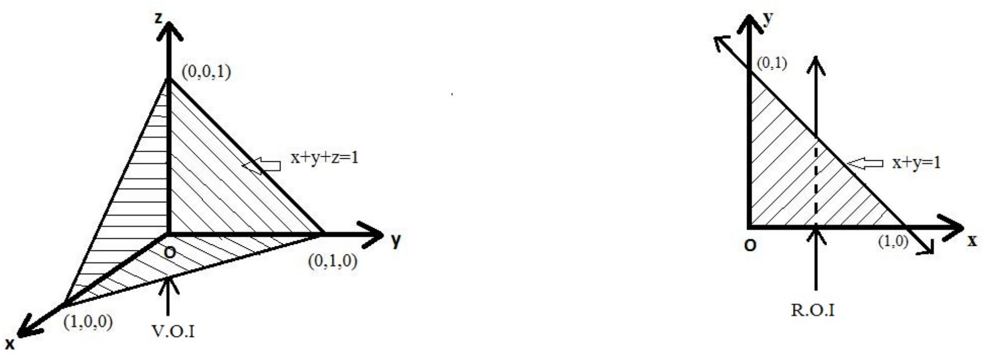


Figure 3:

$\therefore$ Given Integral= $\int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y}\frac{dx\,dy\,dz}{(1+x+y+z)^2}$

$= \int_0^1\int_0^{1-x}\left(\int_0^{1-x-y}(1+x+y+z)^{-2}\,dz\right)\,dy\,dx$

$= \int_0^1\int_0^{1-x}\left[\frac{(1+x+y+z)^{-1}}{-1}\right]\Big|_0^{1-x-y}\,dy\,dx$

$=$ - $\int_0^1\int_0^{1-x}\left[\frac{1}{2}-\frac{1}{1+x+y}\right]\,dy\,dx$

$=$ - $\int_0^1\left[\frac{1}{2}y-log(1+x+y)\right]_0^{1-x}\,dx$

$=$ - $\int_0^1\left[\frac{1}{2}(1-x)-log2+log(1+x)\right]\,dx$

$=\frac{3}{4}-log2$

Triple integrals in cylindrical and spherical co-ordinates

when a calculation in physics , engineering or geometry involves a cylinder, cone or sphere, we can often simplify our work by using cylindrical or spherical co-ordinates. The procedure for transforming to these co-ordinates and evaluating the resulting triple integrals is similar to the transformation to polar co-ordinates in the plane.

Integration in cylindrical co-ordinates
We can obtain cylindrical coordinates for space by combining polar coordinates in the XY plane with the usual z-axis.

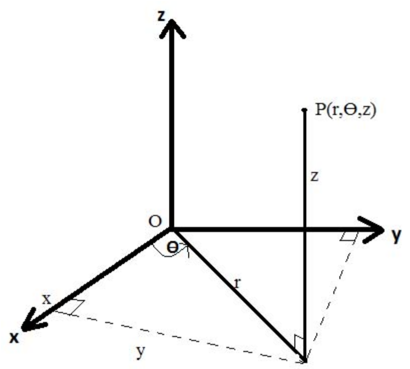


Figure 4:

Equation relating rectangular (x,y,z) and cylindrical coordinates.
$x = r\cos\theta, y = r\sin\theta, z = z, r^2 = x^2 + y^2, \tan\theta = \frac{y}{x}$

0.1 Spherical coordinates and integration

Spherical coordinates locate points in space with two angles and one distance as shown in the figure . The first coordinates $\rho = |\vec{OP}|$ is the point's distance from the origin. Unlike r, the variable 'ρ' is never negative. The second coordinate ϕ is the angle $\vec{OP}$ makes with the positive z axis . It is required to lie in the angle $[0, \pi]$. The third coordinate is as angle θ as measured in the polar coordinates.

Equations relating spherical coordiantes go Cartesian and cylindrical coordinates .
$r = \rho\sin\phi, x = r\cos\theta = \rho\sin\phi\cos\theta, z = \rho\cos\phi, y = r\sin\theta = \rho\sin\phi\sin\theta$

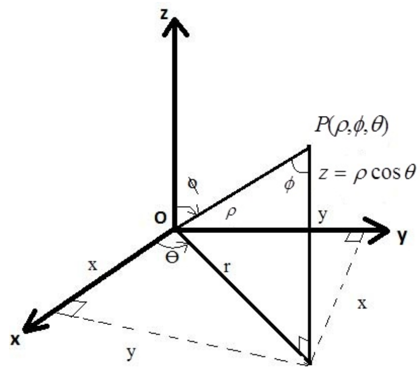


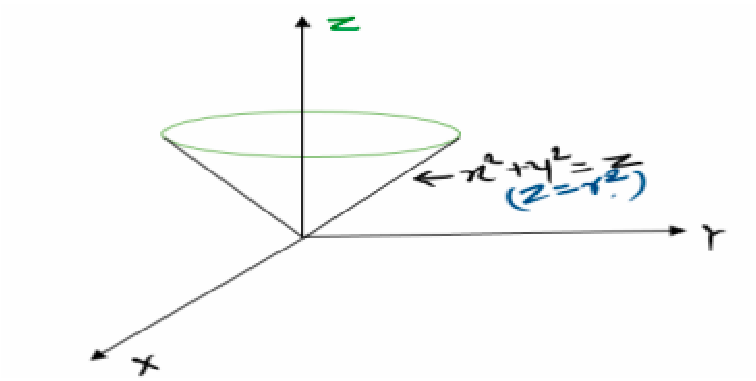
Figure 5:

$\rho = \sqrt{x^2+y^2+z^2} = \sqrt{r^2+z^2}$
$x^2+y^2+z^2 = \rho^2,\ dx\,dy\,dz = \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$

Examples: Triple Integration using Cylindrical Co-ordinates

1. Evaluate $\iiint \sqrt{x^2+y^2}dx\ dy\ dz$ over the volume bounded by the right circular cone $x^2+y^2=z^2,\ z\geq 0,\ and\ \ z=1$
   Solution:Given $I = \iiint \sqrt{x^2+y^2}dx\ dy\ dz$
   We will change the given integral into cylindrical co-ordinates by putting,
   $x = r\cos\theta,\ y = r\sin\theta,\ z = z,\ dxdydz = rdrd\theta dz$
   Then the equation of cone in cylindrical coordinates is

$$x^2+y^2 = z^2$$
$$(r\cos\theta)^2 + (r\sin\theta)^2 = z^2$$
$$r^2 = z^2$$
$$\Rightarrow r^2 = z^2$$



Here, z varies from $z = r\ to\ z = 1$
r varies from $r = 0\ to\ r = 1$
and θ avries from $\theta = 0\ to\ \theta = 2\pi$
Therefore,

$$\begin{aligned}
I &= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \int_{z=r}^{z=1} \sqrt{(r\cos\theta)^2 + (r\sin\theta)^2}\ r\ dz\ dr\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \int_{z=r}^{z=1} r^2\ dz\ dr\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} r^2[z]_r^1\ dr\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} r^2[1-r]\ dr\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi} \left[\frac{r^3}{3} - \frac{r^4}{4}\right]_0^1 \\
&= \frac{1}{12}\int_{\theta=0}^{\theta=2\pi} 1\ d\theta \\
&= \frac{\pi}{6}
\end{aligned}$$

2. Evaluate $I = \iiint z^2 dx\ dy\ dz$ over the volume bounded by the cylinder $x^2 + y^= a^2$ and paraboloid $x^2 + y^2 = z$ and the plane $z = 0$
Solution: $I = \iiint z^2 dx\ dy\ dz$
Change the integral to cylindrical coordinates.
Substitute $x = r\cos\theta$, $y = r\sin\theta,\ \ z = z$
$\therefore dxdydz = rdrd\theta dz$

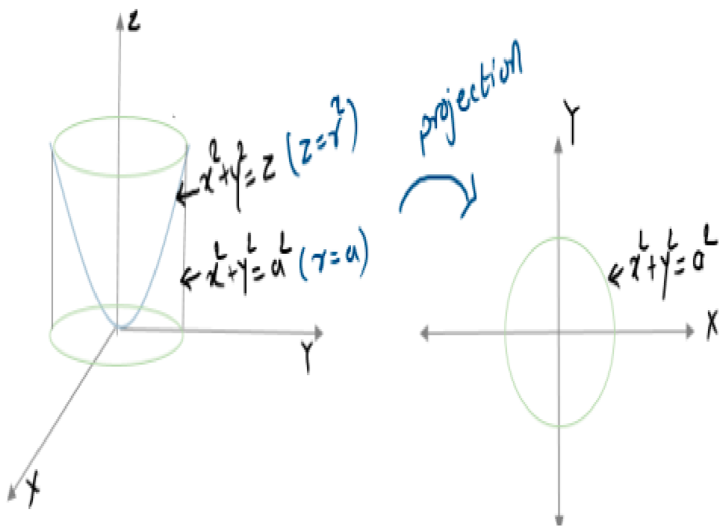
Equation of culinder in cylindrical co-ordinates

$$x^2 + y^2 = a^2$$

$$(r\cos\theta)^2 + (r\sin\theta)^2 = a^2$$
$$\therefore r^2 = a^2$$
$$\Rightarrow r = a$$

Equation of paraboloid

$$x^2 + y^2 = z \Rightarrow r^2 = z$$



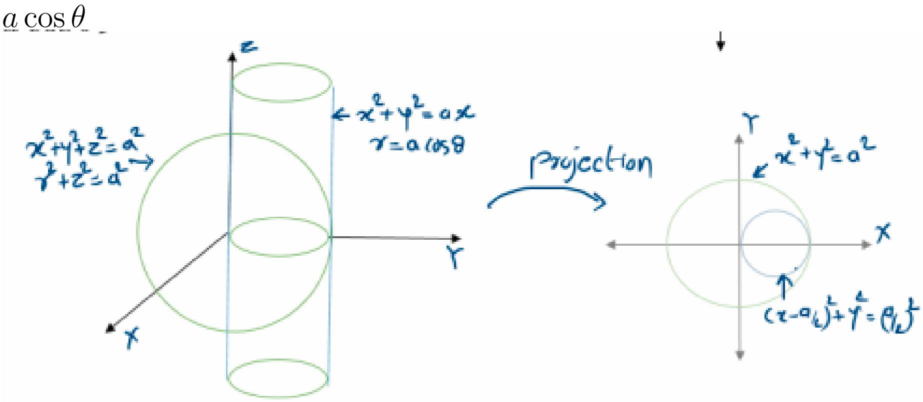
For the given volume
z varies from $z = 0 \ to \ z = r^2$
r varies from $r = 0 \ to \ r = a$
θ varies from $\theta = 0 \ to \ \theta = 2\pi$

$$\begin{aligned}
I &= \int_{\theta=0}^{2\pi} \int_{r=0}^{a} \int_{z=0}^{r^2} z^2 \ r dz dr d\theta \\
&= \int_{\theta=0}^{2\pi} \int_{r=0}^{a} \left[\frac{z^3}{3}\right]_0^{r^2} r \ dr d\theta \\
&= \int_{\theta=0}^{2\pi} \int_{r=0}^{a} \frac{r^7}{3} dr d\theta \\
&= \int_{\theta=0}^{2\pi} \frac{1}{3} \left[\frac{r^8}{8}\right]_0^{a} d\theta \\
&= \frac{a^8}{24} \int_{\theta=0}^{2\pi} 1 \ d\theta \\
&= \frac{\pi a^8}{12}
\end{aligned}$$

3. Evaluate $\iiint z^2 dxdydz$ over the volume common to the sphere $x^2 + y^2 + z^2 = a^2$ and the cylinder $x^2 + y^2 + z^2 = ax$

Substituting $x = r\cos\theta,\ y = r\sin\theta,\ z = z$
and hence $dxdydz = rdrd\theta dz$

Equatio of the sphere is $x^2 + y^2 + z^2 = a^2$ i.e. $r^2 + z^2 = a^2$
and the equation of cylinder is $x^2 + y^2 = ax \ \Rightarrow\ r^2 = ar\cos\theta \ \Rightarrow\ r = a\cos\theta$



The volume of the integration is bounded by sphere and cylinder.
z varies from $z = -\sqrt{a^2 - r^2}$ *and* $z = \sqrt{a^2 - r^2}$
r varies from $r = 0$ *to* $r = a\cos\theta$

θ varies from $\theta = -\frac{\pi}{2}\ to\ \frac{\pi}{2}$

$$I = \int_{\theta=-\pi/2}^{\pi/2} \int_{r=0}^{r=a\cos\theta} \int_{z=-\sqrt{a^2-r^2}}^{\sqrt{a^2-r^2}} z^2\ rdzdrd\theta$$
$$= \int_{\theta=-\pi/2}^{\pi/2} \int_{r=0}^{r=a\cos\theta} \left[\frac{z^3}{3}\right]_{-\sqrt{a^2-r^2}}^{\sqrt{a^2-r^2}} drd\theta$$
$$= \int_{\theta=-\pi/2}^{\pi/2} \int_{r=0}^{r=a\cos\theta} \frac{2}{3}(a^2-r^2)^{3/2} rdrd\theta$$

substitute $a^2 - r^2 = t, \quad \therefore -2rdr = dt \Rightarrow 2rdr = -dt$

$$I = \int_{-\pi/2}^{\pi/2} \frac{1}{3}\left[-\frac{2}{5}(a^2-r^2)^{5/2}\right]_0^{a\cos\theta} d\theta$$
$$= \frac{2a^5}{15}\int_{-\pi/2}^{\pi/2}(1-\sin^5\theta)d\theta$$
$$= \frac{2a^5}{15}\left(\int_{-\pi/2}^{\pi/2} 1\ d\theta - \int_{-\pi/2}^{\pi/2} \sin^5\theta\ d\theta\right)$$
$$= \frac{2a^5}{15}\left([\theta]_{-\pi/2}^{\pi/2} - 0\right) ----as\ \sin^5\theta \text{is an odd function}$$
$$= \frac{2a^5}{15}\left(\frac{\pi}{2}+\frac{\pi}{2}\right)$$
$$= \frac{2a^5}{15}$$

Examples: Triple Integration using spherical Co-ordinates

1. Evaluate $\int\int\int \frac{z^2}{x^2+y^2+z^2}\ dx\ dy\ dz$ over the volume of the sphere $x^2+y^2+z^2=2$.

   Solution: We want to evaluate triple integration over a volume of the sphere, we will change the integral into spherical coordinates by substituting
   $x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$ and $dx\ dy\ dz = \rho^2\sin\phi\ d\rho\ d\theta\ d\phi$.
   The given sphere is $x^2+y^2+z^2=2$
   ρ should vary from $\rho = 0$ to $\rho = \sqrt{2}$
   ϕ should vary from $\phi = 0$ to $\phi = \pi$
   θ should vary from $\theta = 0$ to $\theta = 2\pi$

$$
\begin{aligned}
\therefore \int\int\int \frac{z^2}{x^2+y^2+z^2}\ dx\ dy\ dz &= \int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi}\int_{\rho=0}^{\rho=\sqrt{2}} \frac{\rho^2\cos^2\phi}{\rho^2}\rho^2\sin\phi\ d\rho\ d\phi\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi}\int_{\rho=0}^{\rho=\sqrt{2}} \rho^2\cos^2\phi\sin\phi\ d\rho\ d\phi\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi} \cos^2\phi\sin\phi\left[\frac{\rho^3}{3}\right]_{\rho=0}^{\rho=\sqrt{2}}\ d\phi\ d\theta \\
&= \frac{2\sqrt{2}}{3}\int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi} \cos^2\phi\sin\phi\ d\phi\ d\theta \\
&= \frac{2\sqrt{2}}{3}\int_{\theta=0}^{\theta=2\pi} -\left[\frac{\cos^3\phi}{3}\right]_{\phi=0}^{\phi=\pi}\ d\theta \\
&= \frac{2\sqrt{2}}{3}.\frac{2}{3}\int_{\theta=0}^{\theta=2\pi} 1\ d\theta \\
&= \frac{4\sqrt{2}}{9}2\pi \\
\therefore \int\int\int \frac{z^2}{x^2+y^2+z^2}\ dx\ dy\ dz &= \frac{8\sqrt{2}\pi}{9}
\end{aligned}
$$

2. Evaluate $\int\int\int x^2+y^2+z^2\ dx\ dy\ dz$ over the volume of the First Octant of sphere $x^2+y^2+z^2=a^2$.

   Solution: We want to evaluate triple integration over a volume of the sphere, we will change the integral into spherical coordinates by substituting
   $x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$ and $dx\ dy\ dz = \rho^2\sin\phi\ d\rho\ d\theta\ d\phi$.
   For the First Octant of the sphere $x^2+y^2+z^2=a^2$ i.e $\rho^2=a^2 \implies \rho=a$
   ρ varies from $\rho=0$ to $\rho=a$
   ϕ varies vary from $\phi=0$ to $\phi=\frac{\pi}{2}$
   θ varies vary from $\theta=0$ to $\theta=\frac{\pi}{2}$

$$\begin{aligned}
\therefore \int\int\int x^2+y^2+z^2 \; dx\; dy\; dz &= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{\phi=0}^{\phi=\frac{\pi}{2}} \int_{\rho=0}^{\rho=a} \rho^2.\rho^2 \sin\phi \; d\rho\; d\phi\; d\theta \\
&= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{\phi=0}^{\phi=\frac{\pi}{2}} \left[\frac{\rho^5}{5}\right]_{\rho=0}^{\rho=a} \sin\phi \; d\phi\; d\theta \\
&= \frac{a^5}{5} \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{\phi=0}^{\phi=\frac{\pi}{2}} \sin\phi \; d\phi\; d\theta = \frac{a^5}{5} \int_{\theta=0}^{\theta=\frac{\pi}{2}} -\cos\phi \Big|_{\phi=0}^{\phi=\frac{\pi}{2}} d\theta \\
&= \frac{a^5}{5} \int_{\theta=0}^{\theta=\frac{\pi}{2}} d\theta = \frac{a^5}{5}.\frac{\pi}{2}
\end{aligned}$$

$$\therefore \int\int\int \frac{z^2}{x^2+y^2+z^2} \; dx\; dy\; dz = \frac{a^5\pi}{10}$$

3. Evaluate $\displaystyle\int\int\int \frac{dx\; dy\; dz}{(x^2+y^2+z^2)^{3/2}}$ over the volume bounded by $x^2+y^2+z^2=a^2$ and $x^2+y^2+z^2=b^2$.....$(b>a)$.

   Solution: We want to evaluate triple integration over a volume of the sphere, we will change the integral into spherical coordinates by substituting
   $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, $z = \rho\cos\phi$ and $dx\; dy\; dz = \rho^2\sin\phi\; d\rho\; d\theta\; d\phi$.
   The volume bounded by $x^2+y^2+z^2=a^2 \implies \rho^2=a^2 \implies \rho=a$ and $x^2+y^2+z^2=b^2 \implies \rho^2=b^2 \implies \rho=b$
   ρ varies from $\rho=a$ to $\rho=b$
   ϕ varies vary from $\phi=0$ to π
   θ varies vary from $\theta=0$ to $\theta=2\pi$

$$
\begin{aligned}
\therefore \int\int\int \frac{dx\ dy\ dz}{(x^2+y^2+z^2)^{3/2}} &= \int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi}\int_{\rho=a}^{\rho=b} \frac{\rho^2 \sin\phi\ d\rho\ d\phi\ d\theta}{(\rho^2)^{3/2}} \\
&= \int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi} \sin\phi \int_{\rho=a}^{\rho=b} \frac{1}{\rho}\ d\rho\ d\phi\ d\theta \\
&= \int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi} \sin\phi \quad \log\rho\Big|_{\rho=a}^{\rho=b}\ d\phi\ d\theta \\
&= (\log b - \log a)\int_{\theta=0}^{\theta=2\pi}\int_{\phi=0}^{\phi=\pi} \sin\phi\ d\phi\ d\theta \\
&= \log\frac{b}{a}\int_{\theta=0}^{\theta=2\pi} -\cos\phi\Big|_{\phi=0}^{\phi=\pi}\ d\theta \\
&= \log\frac{b}{a}\int_{\theta=0}^{\theta=2\pi} 2\ d\theta \\
&= 2\log\frac{b}{a}.2\pi \\
\therefore \int\int\int \frac{z^2}{x^2+y^2+z^2}\ dx\ dy\ dz &= 4\pi\log\frac{b}{a}
\end{aligned}
$$

4. Evaluate $\int\int\int \sqrt{1-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)}\ dx\ dy\ dz$ over the volume of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

   Solution: Here V is a volume of ellipsoid, so we will substitute
   $x = a\rho\sin\phi\cos\theta, \quad y = b\rho\sin\phi\sin\theta, \quad z = c\rho\cos\phi$ and $dx\ dy\ dz = abc\ \rho^2\sin\phi\ d\rho\ d\theta\ d\phi$.
   $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} = \frac{(a\rho\sin\phi\cos\theta)^2}{a^2}+\frac{(b\rho\sin\phi\sin\theta)^2}{b^2}+\frac{(c\rho\cos\phi)^2}{c^2} = \rho^2$
   So this substitution will transform ellipsoid into the sphere of radius 1.
   For the volume of ellipsoid in First Octant
   $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} = 1 \implies \rho^2 = 1 \implies \rho = 1$
   Thus, ρ varies from $\rho = 0$ to $\rho = 1$
   ϕ varies vary from $\phi = 0$ to $\pi/2$
   θ varies vary from $\theta = 0$ to $\theta = \pi/2$

$$
\therefore I = 8\int_{\theta=0}^{\theta=\frac{\pi}{2}}\int_{\phi=0}^{\phi=\frac{\pi}{2}}\int_{\rho=0}^{\rho=1} \sqrt{1-\rho^2}\ abc\ \rho^2\sin\phi\ d\rho\ d\phi\ d\theta
$$

Let $\rho = \sin t \implies d\rho = \cos t\ dt$
At $\rho = 0 \implies t = 0$ and $\rho = 1 \implies t = \pi/2$

$$\begin{aligned}
\therefore I &= 8abc \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{\phi=0}^{\phi=\frac{\pi}{2}} \int_{t=0}^{t=\frac{\pi}{2}} \sin^2 t \cos^2 t dt \sin\phi\ d\phi\ d\theta \\
&= 8abc \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{\phi=0}^{\phi=\frac{\pi}{2}} \frac{1}{2}\beta\left(\frac{3}{2}, \frac{3}{2}\right) \sin\phi\ d\phi\ d\theta \\
&= 8abc\ \frac{1}{2}\ \beta\left(\frac{3}{2}, \frac{3}{2}\right) \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{\phi=0}^{\phi=\frac{\pi}{2}} \sin\phi\ d\phi\ d\theta \\
&= 4abc\ \beta\left(\frac{3}{2}, \frac{3}{2}\right) \int_{\theta=0}^{\theta=\frac{\pi}{2}} -\cos\phi\Big|_{\phi=0}^{\phi=\frac{\pi}{2}}\ d\theta \\
&= 4abc\ \beta\left(\frac{3}{2}, \frac{3}{2}\right) \int_{\theta=0}^{\theta=\frac{\pi}{2}} 1\ d\theta
\end{aligned}$$

$$\therefore \int\int\int \sqrt{1 - \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)}\ dx\ dy\ dz = \frac{\pi^2}{4} abc$$